Thermodynamics and Thermochemistry

Thermodynamics: The study of the flow of heat or any other form of energy into or out of a system as it undergoes a physical or chemical transformation is called Thermodynamics.

Scope of Thermodynamics:

(1) Most of the important laws of Physical Chemistry, including the van‟t Hoff law of lowering of vapour pressure, Phase Rule and the Distribution Law, can be derived from the laws of thermodynamics.

(2) It tells whether a particular physical or chemical change can occur under a given set of conditions of temperature, pressure and concentration.

(3) It also helps in predicting how far a physical or chemical change can proceed, until the equilibrium conditions are established.

SYSTEM, BOUNDARY, SURROUNDINGS

A system is that part of the universe which is under thermodynamic study and universe is surroundings.

The real or imaginary surface separating the system from the surrounding boundary.

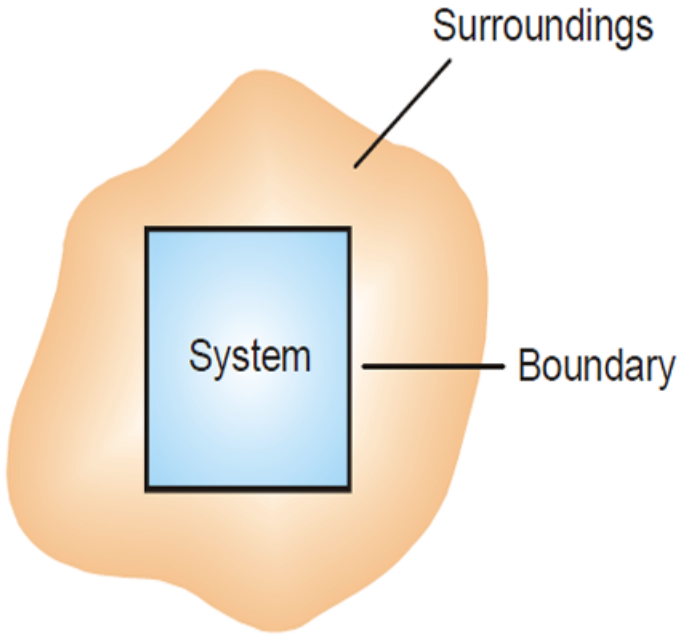


Figure 7.1
Thermodynamic system.

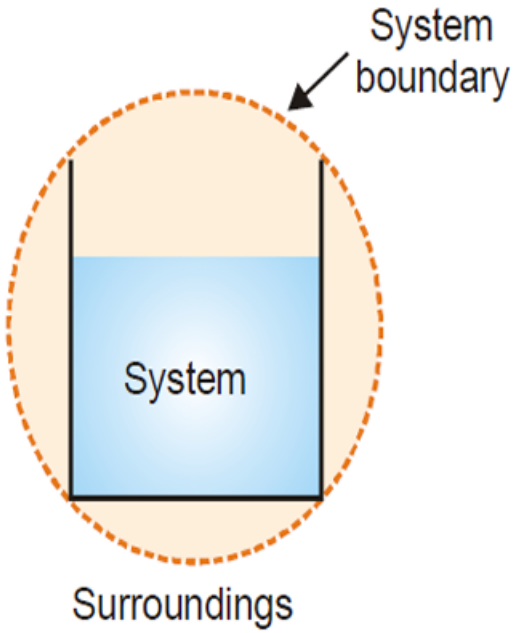


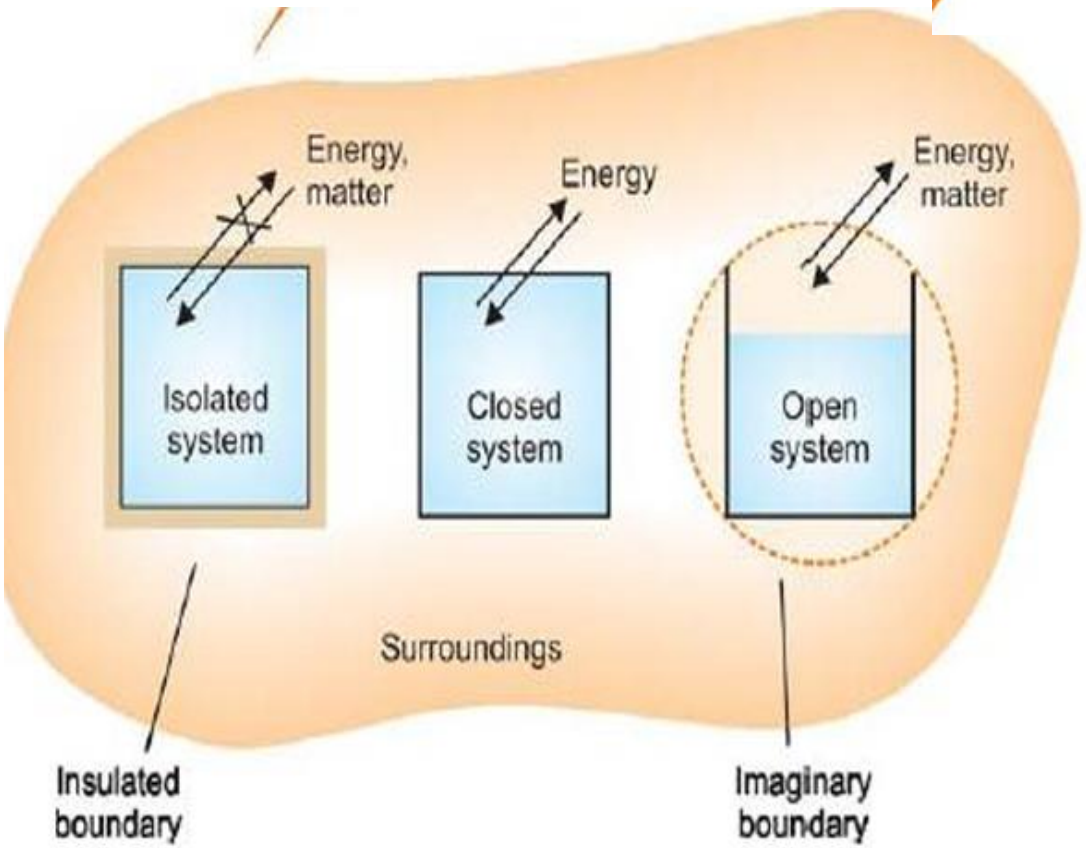
Figure 7.2
Water contained in a beaker constitutes a system.

Types Of Thermodynamic Systems:

(1) Isolated System: When the boundary is both sealed and insulated, no interaction is possible with the surroundings. Therefore, an isolated system is one that can transfer neither matter nor energy to and from its surroundings.

(2) Closed System: Here the boundary is sealed but not insulated. Therefore, a closed system is one which cannot transfer matter but can transfer energy in the form of heat, work and radiation to and from its surroundings.

(3) Open System: In such a system the boundary is open and un-insulated. Therefore, an open system is one which can transfer both energy and matter to and from its surroundings.


Energy,
matter
Energy
Energy,
matter
Isolated
system
Closed
system
Open
system
Surroundings
Insulated
boundary
Imaginary
boundary

Thermodynamic Processes: When a thermodynamic system changes from one state to another, the operation is called a Process. These processes involve the change of conditions (temperature, pressure and volume).

The various types of thermodynamic processes are :

(1) Isothermal Processes: Those processes in which the temperature remains fixed, are termed isothermal processes. This is often achieved by placing the system in a thermostat (a constant temperature bath).For an isothermal process $dT= 0$

(2) Adiabatic Processes: Those processes in which no heat can flow into or out of the system, are called adiabatic processes. Adiabatic conditions can be approached by carrying the process in an insulated container such as „thermos" bottle. High vacuum and highly polished surfaces help to achieve thermal insulation. For an adiabatic process $dq= 0$

(3). Isobaric Processes: Those processes which take place at constant pressure are called isobaric processes. For example, heating of water to its boiling point and its vaporisation take place at the same atmospheric pressure. These changes are, therefore, designated as isobaric processes and are said to take place isobarically. For an isobaric process $dp = 0$

(4). Isochoric Processes: Those processes in which the volume remains constant are known as isochoric processes. The heating of a substance in a non-expanding chamber is an example of isochoric process. For isochoric processes $dV = 0$.

(5). Cyclic Process: When a system in a given state goes through a number of different processes and finally returns to its initial state, the overall process is called a cycle or cyclic process. For a cyclic process $dE = 0$, $dH = 0$.

Reversible And Irreversible Processes: A thermodynamic reverse process is one that takes place infinitesimally slowly and its direction at any point can be reversed by an infinitesimal change in the state of the system.

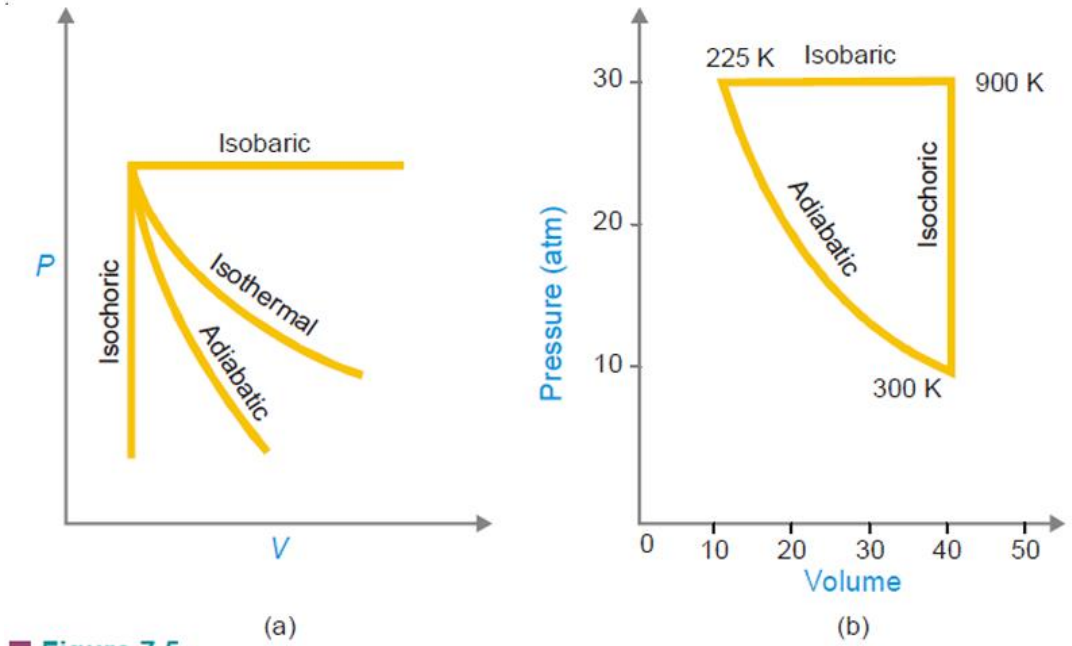


Figure 7.5
(*a*) Four basic thermodynamic processes; (*b*) A cyclic process.

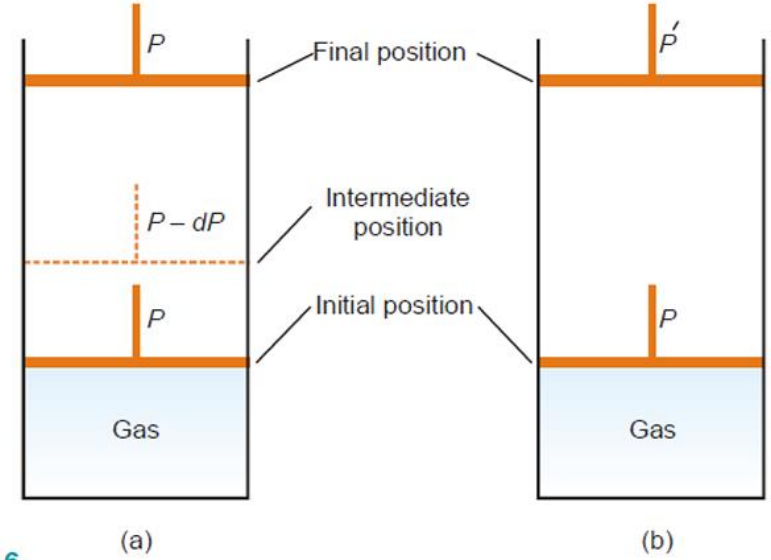


Figure 7.6
(*a*) Reversible expansion occurs by decreasing the pressure on the piston by infinitesimal amounts. (*b*) Irreversible expansion occurs by sudden decrease of pressure from P to P′, when the gas expands rapidly in a single operation.

When a process goes from the initial to the final state in a single step and cannot be carried in the reverse order, it is said to be an irreversible process.

Here the system is in equilibrium state in the beginning and at the end, but not at points in between. Consider a certain quantity of a gas contained in a cylinder having a weightless and frictionless piston. The expansion of the gas can be carried by two methods illustrated in Fig. 7.6. Let the pressure applied to the piston be P and this is equal to the internal pressure of the gas. Since the external and internal pressures are exactly counterbalanced, the piston remains stationary and there is no change in volume of the gas. Now suppose the pressure on the piston is decreased by an infinitesimal amount dP. Thus the external pressure on the piston being P – dP, the piston moves up and the gas will expand by an infinitesimal small amount. The gas will, therefore, be expanded infinitely slowly i.e., by a thermodynamically reversible process. At all stages in the expansion of the gas, dP being negligibly small the gas is maintained in a state of equilibrium throughout. If at any point of the process the pressure is increased by dP, the gas would contract reversibly. On the other hand, the expansion is irreversible (Fig. 7.6 b) if the pressure on the piston is decreased suddenly. It moves upward rapidly in a single operation. The gas is in equilibrium state in the initial and final stages only. The expansion of the gas, in this case, takes place in an irreversible manner.

DIFFERENCES BETWEEN REVERSIBLE AND IRREVERSIBLE PROCESSES

Reversible Process	Irreversible Process
1. It takes place in infinite number of infinitesimally small steps and it would take *infinite time* to occur.	1. It takes place *infinite time*.
2. It is *imaginary* as it assumes the presence of frictionless and weightless piston.	2. It is *real* and can be performed actually.
3. It is in equilibrium state at *all stages* of the operation.	3. It is in equilibrium state only at the *initial and final stages* of the operation.
4. All changes are *reversed* when the process is carried out in reversible direction.	4. After this type of process has occurred all changes *do not return* to the initial state by themselves.
5. It is extremely slow.	5. It proceeds at *measureable speed*.
6. Work done by a reversible process is *greater* than the corresponding irreversible process.	6. Work done by a irreversible process is *smaller* than the corresponding reversible process.

Internal Energy: The total of all the possible kinds of energy of a system, is called its Internal Energy. The word „internal" is often omitted and the energy of a system always implies internal energy. The internal energy of a system, like temperature, pressure, volume, etc., is determined by the state of a system and is independent of the path by which it is obtained. **Hence internal energy of a system is a state function**. For example, we consider the heating of one mole of liquid water from 0º to 100º C. The change in energy is 1.8 kcal and is the same regardless of the form in which this energy is transferred to the water by heating, by performing work, by electrical means or in any other way. The internal energy of a system is represented by the symbol E. It is neither possible nor necessary to calculate the absolute value of internal energy of a system. In thermodynamics we are concerned only with the energy changes when a system changes from one state to another. If ΔE be the difference of energy of the initial state (Ein) and the final state (Ef), we can write $\Delta E = Ef - Ein$

ΔE is +ve if Ef is greater than Ein and –ve if Ef is less than Ein. A system may transfer energy to or from the surroundings as heat or as work, or both.

Laws of Thermodynamics

The Three laws of Thermodynamics

The first laws of thermodynamics, also called conservation of energy. We can use this knowledge to determine the amount of energy in a system, the amount lost as waste heat and the efficiency of the system.

First Law of Thermodynamics: Also known as the law of energy conservation, it states that energy cannot be created or destroyed in an isolated system. The first law is often expressed in the form of the equation

$$\Delta U = Q - W$$

where ΔU is the change in internal energy of the system, Q is the heat added to the system, and W is the work done by the system on its surroundings. This law essentially quantifies the principle of conservation of energy.

The second law of thermodynamics states that the disorder in the universe always increases. As the disorder in the universe increases, the energy is transformed into less usable forms. Thus, the efficiency of any process will always be less than 100%.

The Second Law of Thermodynamics : This law introduces the concept of entropy, a measure of disorder or randomness in a system. The second law states that in an isolated system, the total entropy can never decrease over time.

The third law of thermodynamics tell us that all molecular movement stops at a temperature we call absolute zero, or 0 Kelvin (-273 deg C). Since temperature is a measure of molecular movement, there can be no temperature lower than absolute zero. At this temperature, a perfect crystal has no disorder.

The Third Law of Thermodynamics : A system's entropy approaches a constant value as its temperature approaches absolute zero.

More typically in engineering applications we focus on the 1st law of thermodynamic

First Law Of Thermodynamics: The first law of thermodynamics is, in fact, an application of the broad principle known as the Law of Conservation of Energy to the thermodynamic system. It states that: the total energy of an isolated system remains constant though it may change from one form to another.

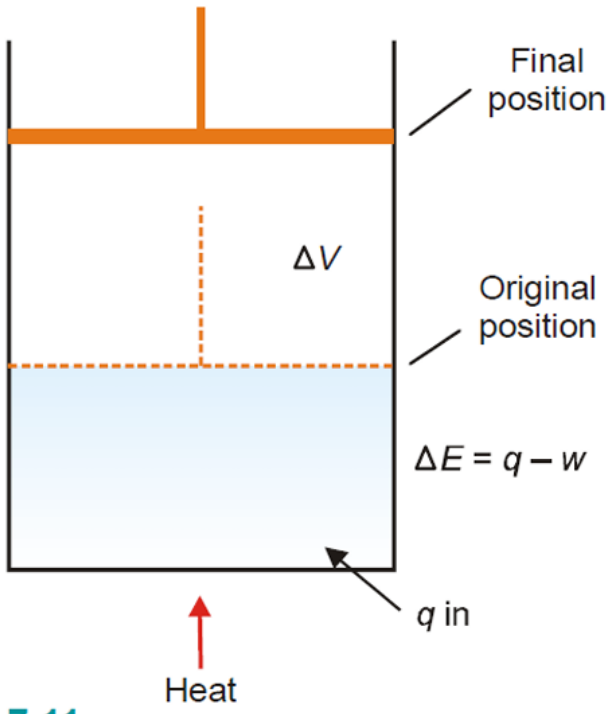


Figure 7.11
Illustration of First Law. Heat adds to internal energy, while work subtracts.

When a system is changed from state A to state B, it undergoes a change in the internal energy from EA to EB.

Thus, we can write $\Delta E = EB - EA$ This energy change is brought about by the evolution or absorption of heat and/or by work being done by the system. Because the total energy of the system must remain constant,

we can write the mathematical statement of the First Law as : $\Delta E = q - w$ ...(1) where q = the amount of heat supplied to the system, w = work done by the system Thus First Law may also be stated as :the net energy change of a closed system is equal to the heat transferred to the system minus the work done by the system. To illustrate the mathematical statement of the First Law, let us consider the system „expanding Hot gas" (Fig. 7.11).

The gas expands against an applied constant pressure by volume ΔV. The total mechanical work done is given by the relation- $w = P \times \Delta V$ ...(2) From (1) and (2), we can restate $\Delta E = q - P \times \Delta V$